

A disproof of Conjecture 00000007679

The zeros of the finite q -binomial sums are real

1 The conjecture

Definition: For the Euler product $(z; q)_\infty = \prod_{k \geq 0} (1 - zq^k)$, the partial sums are $S_N(z) = \sum_{n=0}^N (-1)^n q^{n(n-1)/2} \begin{bmatrix} N \\ n \end{bmatrix}_q z^n$. *Conjecture:* For each $0 < q < 1$, the zero set of $S_N(z)$ converges as N grows to a quasicircle Jordan curve whose shape-parameter envelope is controlled by the two points $z = -q^{-N}$ and $z = q^{-N}$, and the conformal scaling constant of the quasicircle equals $1 + 2q$.

Reading. The conjecture asserts, for each $q \in (0, 1)$, the existence of a quasicircle $\Gamma \subset \mathbb{C}$ such that $Z_N := \{z \in \mathbb{C} : S_N(z) = 0\}$ converges to Γ ; the remaining clauses describe this Γ . A quasicircle is in particular a Jordan curve (see Section 5), so it suffices to show that no Jordan curve is such a limit. We refute the following readings of “ Z_N converges to Γ ”:

- (H) Hausdorff convergence: $d_H(Z_N, \Gamma) \rightarrow 0$ (we use the extended Hausdorff distance, so no finiteness convention is involved);
- (W) the weakest requirement shared by all usual notions of set convergence in $\mathbb{C}$ (Hausdorff, Kuratowski lower or upper limit, local convergence on compacta): every point of Γ is a limit, or a cluster point, of zeros, i.e. $\Gamma \subseteq \overline{\bigcup_N Z_N}$.

Both are refuted for every $q \in (0, 1)$, and also after any real affine normalisation $Z_N \mapsto a_N Z_N + b_N$ ($a_N, b_N \in \mathbb{R}$; this includes the rescaling $z \mapsto q^N z$ suggested by the points $\pm q^{-N}$). Not formalised: convergence on the Riemann sphere; there the closure of $\bigcup_N Z_N$ is the countable set $\{q^{-k} : k \geq 0\} \cup \{\infty\}$, which also contains no Jordan curve (a Jordan curve is uncountable), but this is argued only in prose here. Normalisations by non-real complex factors are not treated.

2 Definitions (as in the Lean file)

Definition 1. $(q; q)_n = \prod_{i=0}^{n-1} (1 - q^{i+1})$ (`qPoch`). The Gaussian binomial coefficient is $\begin{bmatrix} N \\ k \end{bmatrix}_q = (q; q)_N / ((q; q)_k (q; q)_{N-k})$ for $k \leq N$ and 0 for $k > N$ (`gaussBinom`). This is the textbook formula $\frac{(1-q^N) \cdots (1-q^{N-k+1})}{(1-q) \cdots (1-q^k)}$ after cancelling $(q; q)_{N-k}$. $S_N(z)$ is defined literally as in the conjecture, with the exponent $n(n-1)/2$ in $\mathbb{N}$ (`S`); Z_N is `zeroSet q N`. A Jordan curve is the image of a continuous injective map from the unit circle into $\mathbb{C}$ (`IsJordanCurve`, using Mathlib’s `Circle`).

3 Proof

Lemma 2 (*q*-Pascal rule, gaussBinom_pascal). For $0 < q < 1$ and all $N, k \geq 0$: $\begin{bmatrix} N+1 \\ k+1 \end{bmatrix}_q =$

$$\begin{bmatrix} N \\ k+1 \end{bmatrix}_q + q^{N-k} \begin{bmatrix} N \\ k \end{bmatrix}_q.$$

Proof. For $k > N$ all three terms vanish; for $k = N$ both sides equal 1. For $k < N$ write $N = k + 1 + m$; after clearing the (nonzero) denominators the identity becomes $1 - q^{N+1} = (1 - q^{m+1}) + q^{m+1}(1 - q^{k+1})$, which holds since $N + 1 = k + m + 2$. $\square$

Lemma 3 (finite *q*-binomial theorem, S_succ, S_eq_prod). $S_{N+1}(z) = (1 - zq^N)S_N(z)$, hence $S_N(z) = \prod_{k=0}^{N-1} (1 - zq^k)$.

Proof. Let $t_N(n) = (-1)^n q^{n(n-1)/2} \begin{bmatrix} N \\ n \end{bmatrix}_q z^n$, which vanishes for $n > N$. By the *q*-Pascal rule and $(m+1)m/2 + (N-m) = N + m(m-1)/2$ for $m \leq N$, we get $t_{N+1}(m+1) = t_N(m+1) - zq^N t_N(m)$ for all m . Summing over m and using $t_N(0) = t_{N+1}(0) = 1$ gives the recursion; induction from $S_0 = 1$ gives the product. (This is the Cauchy binomial theorem with $t = -z$.) $\square$

Hence $Z_N = \{q^{-k} : 0 \leq k < N\} \subset \mathbb{R}$ (zeroSet_eq).

Lemma 4 (jordan_not_subset_real). No Jordan curve is contained in the real line.

Proof. Let γ be continuous and injective on the circle with image in $\mathbb{R}$, and $f(t) = \operatorname{Re} \gamma(e^{it})$. Then f is continuous, $f(2\pi) = f(0)$, and f is injective on $[0, 2\pi)$ (because γ is injective, its values are real, and $t \mapsto e^{it}$ is injective on $[0, 2\pi)$). So $f(0) \neq f(\pi)$; let $c = (f(0) + f(\pi))/2$. The intermediate value theorem on $[0, \pi]$ and on $[\pi, 2\pi]$ gives $s \in (0, \pi)$ and $t \in (\pi, 2\pi)$ with $f(s) = f(t) = c$, contradicting injectivity. $\square$

Lemma 5 (subset_closure_of_tendsto). If $d_H(Y_N, \Gamma) \rightarrow 0$ then $\Gamma \subseteq \overline{\bigcup_N Y_N}$.

Proof. For $x \in \Gamma$, $\operatorname{dist}(x, \bigcup_N Y_N) \leq \operatorname{dist}(x, Y_N) \leq d_H(Y_N, \Gamma)$ for every N . $\square$

Theorem 6 (conjecture_7679_false). For every $q \in (0, 1)$: (1) $S_N(z) = \prod_{k < N} (1 - zq^k)$ for all N, z ; (2) $Z_N = \{q^{-k} : k < N\}$; (3) for all real sequences a_N, b_N , no Jordan curve Γ satisfies $\Gamma \subseteq \bigcup_N (a_N Z_N + b_N)$; (4) for all real sequences a_N, b_N , there is no Jordan curve Γ with $d_H(a_N Z_N + b_N, \Gamma) \rightarrow 0$.

Proof. All sets $a_N Z_N + b_N$ lie in the closed set $\{\operatorname{Im} z = 0\}$, hence so does the closure of their union; apply the two preceding lemmas. $\square$

With $a_N = 1$, $b_N = 0$ this refutes readings (H) and (W), so the zero sets do not converge to any Jordan curve, and in particular not to a quasicircle. The clauses about the envelope and the scaling constant $1 + 2q$ concern a curve that does not exist.

4 Lean formalisation and axiom audit

File `lean/Conjecture7679/Basic.lean` (Lean 4 v4.33.1, Mathlib v4.33.1, only import `Mathlib`). Main theorem: `C7679.conjecture_7679_false`. Supporting results: `gaussBinom_pascal`, `term_succ`, `S_succ`, `S_eq_prod`, `zeroSet_eq`, `zeroSet_real`, `jordan_not_subset_real`, `subset_closure_of_tendsto`. Hausdorff distance is Mathlib's `Metric.hausdorffEDist`. The output of `#print axioms C7679.conjecture_7679_false` is `[propext, Classical.choice, Quot.sound]`. There is no `sorry` and no `native_decide`.

5 Sources

- Wikipedia, *Quasicircle*, <https://en.wikipedia.org/wiki/Quasicircle>: “a quasicircle is a Jordan curve in the complex plane that is the image of a circle under a quasiconformal mapping of the plane onto itself.”
- Wikipedia, *Jordan curve theorem*, https://en.wikipedia.org/wiki/Jordan_curve_theorem: “A Jordan curve or a simple closed curve in the plane is the image of an injective continuous map of a circle into the plane.”
- Wikipedia, *Gaussian binomial coefficient*, https://en.wikipedia.org/wiki/Gaussian_binomial_coefficient: the definition $\binom{m}{r}_q = \frac{(1-q^m)(1-q^{m-1})\dots(1-q^{m-r+1})}{(1-q)(1-q^2)\dots(1-q^r)}$, “If $r > m$, this evaluates to 0”, and the Cauchy binomial theorem $\prod_{k=0}^{n-1}(1+q^k t) = \sum_{k=0}^n q^{k(k-1)/2} \binom{n}{k}_q t^k$. The identity is reproved in Lean; the source is only for the definitions.